

2. Find the general solution of the system

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= 0 \\ -2x_1 + 4x_2 + 5x_3 - 5x_4 &= 3 \\ 3x_1 - 6x_2 - 6x_3 + 8x_4 &= 2 \end{aligned}$$

Soln: Augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ -2 & 4 & 5 & -5 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array} \right]$$

Row 2 = Row 2 + 2 × Row 1
(Step 1)

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array} \right]$$

Step 2: Row 3 = Row 3 - 3 × Row 1

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 0 & 0 & -3 & -1 & 2 \end{array} \right]$$

Step 3: Row 3 = Row 3 + Row 2

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 0 & 0 & 0 & 0 & 5 \end{array} \right]$$

From Theorem 2, the system is inconsistent.

3. Suppose a 4×7 coefficient matrix for a system of equations has 4 pivots. Is the system consistent? If the system is consistent, how many solutions are there?

Soln: The variables corresponding to the pivot positions are called basic variables. $\therefore$ There are 4 basic variables. The variables corresponding to the columns not containing the pivots are called free variables. $\therefore$ There are 3 free variables. In the reduced row echelon form, each equation has only one basic variable. Since there are 4 equations, there should be 4 basic variables which is actually the case. $\therefore$ The system is consistent and since there are at least one free variable, there are infinitely many solutions.